# Reproductive Health

Reproductive health simply refers to healthy reproductive organs with normal functions, but the term has a broader perspective and includes the emotional and social aspects of reproduction also.

## What the WHO means by it

- according to the World Health Organisation (WHO), reproductive health means a total well-being in all aspects of reproduction, that is physical, emotional, behavioural and social
- so a society whose people have physically and functionally normal reproductive organs, and normal emotional and behavioural interactions among them in all sex-related aspects, might be called reproductively healthy

## Where India began

- India was amongst the first countries in the world to initiate action plans and programmes at a national level to attain total reproductive health as a social goal
- these programmes, called family planning, were initiated in 1951 and were periodically assessed over the past decades
- improved programmes covering wider reproduction-related areas are currently in operation

## In this chapter

3.1 Reproductive health
3.4 Population stabilisation
3.5 Contraceptive methods
3.9 Choosing a method
3.10 Medical termination of pregnancy
3.13 Infertility

## 3.1 Awareness and education

### The RCH programmes

- the current programmes run under the popular name Reproductive and Child Health Care (RCH) programmes
- creating awareness among people about various reproduction related aspects, and providing facilities and support for building up a reproductively healthy society, are the major tasks under them
- with the help of audio-visual and print media, governmental and non-governmental agencies have taken various steps to create that awareness, and parents, other close relatives, teachers and friends also have a major role in the dissemination of this information

### Sex education

- introduction of sex education in schools should be encouraged, to provide right information to the young and so discourage children from believing in myths and having misconceptions about sex-related aspects
- proper information about reproductive organs, adolescence and related changes, safe and hygienic sexual practices, sexually transmitted diseases (STD) and AIDS would help people, especially those in the adolescent age group, to lead a reproductively healthy life
- educating people, especially fertile couples and those of marriageable age, about available birth control options, care of pregnant mothers, post-natal care of mother and child, the importance of breast feeding and equal opportunities for the male and the female child would address the importance of bringing up socially conscious healthy families of desired size
- awareness of the problems due to uncontrolled population growth, and of social evils such as sex-abuse and sex-related crimes, needs to be created so that people can think and take up necessary steps to prevent them

## 3.2 Infrastructure and research

### What implementation needs

- successful implementation of the various action plans requires strong infrastructural facilities, professional expertise and material support, all essential to provide medical assistance and care in reproduction-related problems such as pregnancy, delivery, STDs, abortions, contraception, menstrual problems and infertility
- better techniques and new strategies have to be implemented from time to time to provide more efficient care and assistance
- the statutory ban on amniocentesis for sex-determination, to legally check the increasing menace of female foeticides, and massive child immunisation are some programmes that merit mention
- in amniocentesis some of the amniotic fluid of the developing foetus is taken to analyse the fetal cells and dissolved substances, and the procedure is used to test for the presence of certain genetic disorders such as down syndrome, haemoplilia and sickle-cell anemia, and to determine the survivability of the foetus

### Research, and the signs of improvement

- research on various reproduction-related areas is encouraged and supported by governmental and non-governmental agencies, to find new methods and to improve upon existing ones, Saheli, a new oral contraceptive for females, having been developed by scientists at the Central Drug Research Institute (CDRI) in Lucknow, India
- better awareness about sex related matters, an increased number of medically assisted deliveries and better post-natal care leading to decreased maternal and infant mortality rates, an increased number of couples with small families, better detection and cure of STDs and overall increased medical facilities for all sex-related problems all indicate improved reproductive health of the society

## 3.3 Population stabilisation and birth control

### The numbers

- in the last century an all-round development in various fields significantly improved the quality of life of the people, but increased health facilities along with better living conditions had an explosive impact on the growth of population
- the world population, which was around 2 billion, that is 2000 million, in 1900, rocketed to about 6 billion by 2000 and 7.2 billion in 2011
- a similar trend was observed in India, whose population, approximately 350 million at the time of independence, reached close to the billion mark by 2000 and crossed 1.2 billion in May 2011, a rapid decline in death rate, maternal mortality rate (MMR) and infant mortality rate (IMR) as well as an increase in the number of people of reproducible age being probable reasons
- through the Reproductive Child Health (RCH) programme the population growth rate was brought down only marginally, and according to the 2011 census report it was less than 2 per cent, that is 20/1000/year, a rate at which the population could still increase rapidly, which could lead to an absolute scarcity of even the basic requirements of food, shelter and clothing

### The measures taken

- the most important step is to motivate smaller families by using various contraceptive methods, and advertisements, posters and bills show a happy couple with two children under the slogan Hum Do Hamare Do, that is we two, our two
- many couples, mostly the young, urban, working ones, have even adopted a one child norm
- statutory raising of the marriageable age of the female to 18 years and of males to 21 years, and incentives given to couples with small families, are two of the other measures
- an ideal contraceptive should be user-friendly, easily available, effective and reversible with no or least side-effects, and should in no way interfere with the sexual drive, desire or the sexual act of the user, and the wide range presently available groups broadly into natural or traditional, barrier, IUDs, oral contraceptives, injectables, implants and surgical methods

**A sheath of thin rubber or latex**

**Worn to cover the penis just before coitus**

**Nirodh**, a popular brand for the male

**Figure 3.A** Condom for male

**The same barrier principle for the female**

**Covering the vagina and cervix**

**Disposable**, and can be self-inserted

**Figure 3.B** Condom for female

**An intra uterine device**, inserted through the vagina

**Copper ions released within the uterus**

**Sperm motility and fertilising capacity suppressed**

**Figure 3.C** Copper T (CuT)

## 3.4 Natural and barrier methods

### Natural methods

- these work on the principle of avoiding chances of ovum and sperms meeting
- periodic abstinence is one such method, in which couples avoid coitus from day 10 to 17 of the menstrual cycle, when ovulation could be expected and chances of fertilisation are very high, which is why it is called the fertile period
- withdrawal or coitus interruptus is another method, in which the male partner withdraws his penis from the vagina just before ejaculation so as to avoid insemination
- lactational amenorrhea, that is absence of menstruation, is based on the fact that ovulation and therefore the cycle do not occur during the period of intense lactation following parturition, so as long as the mother breast-feeds the child fully the chances of conception are almost nil, though this has been reported to be effective only upto a maximum of six months after parturition, and as no medicines or devices are used side effects are almost nil while chances of failure are high

### Barrier methods

- in these, ovum and sperms are prevented from physically meeting with the help of barriers, and such methods are available for both males and females
- condoms are barriers made of thin rubber or latex sheath used to cover the penis in the male or the vagina and cervix in the female, just before coitus, so that the ejaculated semen would not enter the female reproductive tract, and their use has increased in recent years due to the additional benefit of protecting the user from contracting STIs and AIDS
- diaphragms, cervical caps and vaults are also barriers made of rubber, inserted into the female reproductive tract to cover the cervix during coitus, preventing conception by blocking the entry of sperms through the cervix, and they are reusable
- spermicidal creams, jellies and foams are usually used along with these barriers to increase their contraceptive efficiency

## 3.5 Intra uterine devices

### How IUDs work

- these devices are inserted by doctors or expert nurses in the uterus through the vagina
- they are presently available as non-medicated IUDs such as the Lippes loop, copper releasing IUDs such as CuT, Cu7 and Multiload 375, and hormone releasing IUDs such as Progestasert and LNG-20
- IUDs increase phagocytosis of sperms within the uterus, and the Cu ions released suppress sperm motility and the fertilising capacity of sperms

**Progestogens**, alone or with estrogen

**Placed under the skin as an implant**

**Acting like the pills**, but for much longer

**Figure 3.D** Implants

**A small part of the vas deferens removed or tied**

**Through a small incision on the scrotum**

**Gamete transport blocked in the male**

**Figure 3.E** Vasectomy

**A small part of the fallopian tube removed or tied**

**Through a small incision in the abdomen or the vagina**

**Gamete transport blocked in the female**

**Figure 3.F** Tubectomy

## 3.6 Pills, injectables and implants

### The pills

- the hormone releasing IUDs, in addition, make the uterus unsuitable for implantation and the cervix hostile to the sperms, and IUDs are ideal contraceptives for females who want to delay pregnancy or space children, being one of the most widely accepted methods in India
- oral administration of small doses of either progestogens or progestogen and estrogen combinations is another method used by females, in the form of tablets, hence popularly called the pills
- pills have to be taken daily for a period of 21 days, starting preferably within the first five days of the menstrual cycle, and after a gap of 7 days, during which menstruation occurs, repeated in the same pattern till the female desires to prevent conception
- they inhibit ovulation and implantation and alter the quality of cervical mucus to prevent or retard the entry of sperms, are very effective with lesser side effects and are well accepted, while Saheli, the new oral contraceptive, contains a non-steroidal preparation and is a once a week pill with very few side effects and high contraceptive value

### Injectables, implants and emergency use

- progestogens alone or in combination with estrogen can also be used by females as injections or implants under the skin
- their mode of action is similar to that of pills and their effective periods are much longer
- administration of progestogens, or progestogen and estrogen combinations, or IUDs within 72 hours of coitus has been found to be very effective as emergency contraception, to avoid possible pregnancy due to rape or casual unprotected intercourse

## 3.7 Surgical methods

### Sterilisation

- surgical methods, also called sterilisation, are generally advised for the male or female partner as a terminal method to prevent any more pregnancies
- surgical intervention blocks gamete transport and thereby prevents conception
- the sterilisation procedure in the male is called vasectomy and that in the female tubectomy

### The two operations

- in vasectomy a small part of the vas deferens is removed or tied up through a small incision on the scrotum
- in tubectomy a small part of the fallopian tube is removed or tied up through a small incision in the abdomen or through the vagina
- these techniques are highly effective but their reversibility is very poor

### Choosing a method

- the selection of a suitable contraceptive method and its use should always be undertaken in consultation with qualified medical professionals
- contraceptives are not regular requirements for the maintenance of reproductive health, and are in fact practiced against a natural reproductive event, that is conception or pregnancy
- one is forced to use these methods either to prevent pregnancy or to delay or space pregnancy due to personal reasons, and their widespread use has a significant role in checking uncontrolled growth of population
- their possible ill-effects, such as nausea, abdominal pain, breakthrough bleeding, irregular menstrual bleeding or even breast cancer, though not very significant, should not be totally ignored

## 3.8 Medical termination of pregnancy

### What MTP is

- intentional or voluntary termination of pregnancy before full term is called medical termination of pregnancy (MTP) or induced abortion
- nearly 45 to 50 million MTPs are performed in a year all over the world, which accounts for 1/5th of the total number of conceived pregnancies in a year
- whether to accept or legalise MTP is being debated in many countries, due to the emotional, ethical, religious and social issues involved, and the Government of India legalised it in 1971 with some strict conditions to avoid its misuse, such restrictions being all the more important to check the indiscriminate and illegal female foeticides reported to be high in India
- MTP is done to get rid of unwanted pregnancies, either from casual unprotected intercourse, or failure of the contraceptive used during coitus, or rape, and is also essential in certain cases where continuation of the pregnancy could be harmful or even fatal to the mother or the foetus or both

### The Amendment Act of 2017

- the Medical Termination of Pregnancy (Amendment) Act, 2017 was enacted by the government of India with the intension of reducing the incidence of illegal abortion and consequent maternal mortality and morbidity
- under this Act a pregnancy may be terminated on certain considered grounds within the first 12 weeks of pregnancy on the opinion of one registered medical practitioner, while if the pregnancy has lasted more than 12 weeks but fewer than 24 weeks, two registered medical practitioners must be of the opinion, formed in good faith, that the required ground exists
- the first ground is that the continuation of the pregnancy would involve a risk to the life of the pregnant woman, or grave injury to her physical or mental health
- the second is that there is a substantial risk that if the child were born it would suffer from such physical or mental abnormalities as to be seriously handicapped

## 3.9 Risks and misuse

### When MTP is safe, and when it is abused

- MTPs are considered relatively safe during the first trimester, that is upto 12 weeks of pregnancy, while second trimester abortions are much riskier
- one disturbing trend is that a majority of MTPs are performed illegally by unqualified quacks, which is not only unsafe but could be fatal
- another dangerous trend is the misuse of amniocentesis to determine the sex of the unborn child, frequently followed, if the foetus is found to be female, by MTP, which is totally against what is legal, and such practices should be avoided because they are dangerous both for the young mother and the foetus
- effective counselling on the need to avoid unprotected coitus and on the risk factors involved in illegal abortions, as well as providing more health care facilities, could reverse this unhealthy trend

## 3.10 Sexually transmitted infections

### What they are

- infections or diseases transmitted through sexual intercourse are collectively called sexually transmitted infections (STI), or venereal diseases (VD), or reproductive tract infections (RTI)
- gonorrhoea, syphilis, genital herpes, chlamydiasis, genital warts, trichomoniasis, hepatitis-B and HIV leading to AIDS are some of the common ones, HIV infection being the most dangerous
- some of these, such as hepatitis-B and HIV, can also be transmitted by sharing injection needles and surgical instruments with infected persons, by transfusion of blood, or from an infected mother to the foetus
- except for hepatitis-B, genital herpes and HIV infections, the other diseases are completely curable if detected early and treated properly

### Why they go undetected

- early symptoms of most of these are minor and include itching, fluid discharge, slight pain and swellings in the genital region
- infected females may often be asymptomatic and hence may remain undetected for long, and the absence or insignificance of symptoms in the early stages, together with the social stigma attached to STIs, deters infected persons from going for timely detection and proper treatment
- this could lead to later complications, including pelvic inflammatory diseases (PID), abortions, still births, ectopic pregnancies, infertility or even cancer of the reproductive tract, so STIs are a major threat to a healthy society and their prevention or early detection and cure is given prime consideration under the reproductive health-care programmes
- though all persons are vulnerable, incidences are reported to be very high among persons in the age group of 15-24 years

### How to stay free of them

- avoid sex with unknown partners or multiple partners
- always try to use condoms during coitus
- in case of doubt, go to a qualified doctor for early detection and get complete treatment if diagnosed with infection

## 3.11 Infertility

### Why couples are childless

- a large number of couples all over the world, including India, are infertile, that is unable to produce children in spite of unprotected sexual co-habitation
- the reasons could be many, physical, congenital, diseases, drugs, immunological or even psychological
- in India the female is often blamed for the couple being childless, but more often than not the problem lies in the male partner
- specialised health care units such as infertility clinics could help in diagnosis and corrective treatment of some of these disorders, and where such corrections are not possible the couples could be assisted through certain special techniques commonly known as assisted reproductive technologies (ART)

## 3.12 Assisted reproductive technologies

### IVF and embryo transfer

- in vitro fertilisation, that is fertilisation outside the body in almost the same conditions as in the body, followed by embryo transfer (ET) is one such method, popularly known as the test tube baby programme
- in it, ova from the wife or a donor and sperms from the husband or a donor are collected and induced to form a zygote under simulated conditions in the laboratory
- the zygote or early embryos with upto 8 blastomeres could then be transferred into the fallopian tube, that is zygote intra fallopian transfer or ZIFT, and embryos with more than 8 blastomeres into the uterus, that is intra uterine transfer or IUT, to complete further development
- embryos formed by in-vivo fertilisation, that is fusion of gametes within the female, could also be used for such transfer, to assist females who cannot conceive

### GIFT, ICSI and AI

- transfer of an ovum collected from a donor into the fallopian tube, that is gamete intra fallopian transfer or GIFT, of another female who cannot produce one but can provide a suitable environment for fertilisation and further development, is another method attempted
- intra cytoplasmic sperm injection (ICSI) is a specialised procedure to form an embryo in the laboratory, in which a sperm is directly injected into the ovum
- infertility cases due either to the inability of the male partner to inseminate the female or to very low sperm counts in the ejaculates could be corrected by artificial insemination (AI), in which semen collected either from the husband or a healthy donor is artificially introduced into the vagina or into the uterus, that is intra-uterine insemination or IUI

### Why adoption still matters

- though the options are many, all these techniques require extremely high precision handling by specialised professionals and expensive instrumentation, so the facilities are presently available only in very few centres in the country and their benefits are affordable to only a limited number of people
- emotional, religious and social factors are also deterrents in the adoption of these methods
- since the ultimate aim of all these procedures is to have children, and in India there are so many orphaned and destitute children who would probably not survive till maturity unless taken care of, and our laws permit legal adoption, that is as yet one of the best methods for couples looking for parenthood

## 3.13 Summary

### What reproductive health means

- reproductive health refers to a total well-being in all aspects of reproduction, that is physical, emotional, behavioural and social
- our nation was the first in the world to initiate various action plans at national level towards attaining a reproductively healthy society
- counselling and creating awareness among people about reproductive organs, adolescence and associated changes, safe and hygienic sexual practices and sexually transmitted infections including AIDS is the primary step towards reproductive health
- providing medical facilities and care for problems such as menstrual irregularities, pregnancy related aspects, delivery, medical termination of pregnancy, STIs, birth control, infertility and post natal child and maternal management is another important aspect of the Reproductive and Child Health Care programmes

### Population and contraception

- an overall improvement in reproductive health has taken place in our country, as indicated by reduced maternal and infant mortality rates, early detection and cure of STIs and assistance to infertile couples
- improved health facilities and better living conditions promoted an explosive growth of population, and such growth necessitated intense propagation of contraceptive methods
- various contraceptive options are available now, such as natural, traditional, barrier, IUDs, pills, injectables, implants and surgical methods
- though contraceptives are not regular requirements for reproductive health, one is forced to use them to avoid pregnancy or to delay or space pregnancy

### MTP, STIs and infertility

- medical termination of pregnancy is legalised in our country, and MTP is generally performed to get rid of unwanted pregnancy due to rapes or casual relationships, and also in cases where the continuation of pregnancy could be harmful or even fatal to the mother, the foetus or both
- infections or diseases transmitted through sexual intercourse are called sexually transmitted diseases, and pelvic inflammatory diseases (PIDs), still birth and infertility are some of their complications, while early detection facilitates better cure, and avoiding sexual intercourse with unknown or multiple partners and using condoms during coitus are simple precautions
- inability to conceive or produce children even after 2 years of unprotected sexual cohabitation is called infertility, and various methods are now available to help such couples, in vitro fertilisation followed by transfer of the embryo into the female genital tract being one such method, commonly known as the Test Tube Baby Programme

## 3.14 Exercises

### From this chapter

1. What do you think is the significance of reproductive health in a society?
2. Suggest the aspects of reproductive health which need to be given special attention in the present scenario.
3. Is sex education necessary in schools? Why?
4. Do you think that reproductive health in our country has improved in the past 50 years? If yes, mention some such areas of improvement.
5. What are the suggested reasons for population explosion?
6. Is the use of contraceptives justified? Give reasons.
7. Removal of gonads cannot be considered as a contraceptive option. Why?
8. Amniocentesis for sex determination is banned in our country. Is this ban necessary? Comment.
9. Suggest some methods to assist infertile couples to have children.
10. What are the measures one has to take to prevent from contracting STDs?
11. State True/False with explanation (a) Abortions could happen spontaneously too. (True/False) (b) Infertility is defined as the inability to produce a viable offspring and is always due to abnormalities/defects in the female partner. (True/False) (c) Complete lactation could help as a natural method of contraception. (True/False) (d) Creating awareness about sex related aspects is an effective method to improve reproductive health of the people. (True/False)
12. Correct the following statements : (a) Surgical methods of contraception prevent gamete formation. (b) All sexually transmitted diseases are completely curable. (c) Oral pills are very popular contraceptives among the rural women. (d) In E. T. techniques, embryos are always transferred into the uterus.